

Lab 22 – Advanced Ethical Considerations: Addressing Complex Scenarios and Limitations

Task 1: Bias Detection in AI Suggestions

Prompt:

1. Create a Python program to shortlist candidates based on skills and experience.
2. Add conditions to select the best candidate using marks and experience.
3. Modify the logic to also ensure diversity among selected candidates.
4. Compare both results and check if selection changes.

Code:

```
somes > lab22.py > ...
1  candidates = [
2      ("Alice", 85, 2),
3      ("Bob", 90, 3),
4      ("Charlie", 80, 5)
5  ]
6
7  # Best candidate (only marks)
8  best = max(candidates, key=lambda x: x[1])
9  print("Best Candidate:", best)
10
11 # Diversity logic (experience priority)
12 diverse = max(candidates, key=lambda x: x[2])
13 print("Diverse Selection:", diverse)
```

Output:

```
PS C:\Users\rahul\Desktop\AI Assistanting coding> & C:\Users\rahul\AI Assistanting coding\somes\lab22.py
Best Candidate: ('Bob', 90, 3)
Diverse Selection: ('Charlie', 80, 5)
PS C:\Users\rahul\Desktop\AI Assistanting coding>
```

Explanation:

1. Program selects candidates based on different conditions.
2. First logic favors marks while second favors experience.
3. Results show how selection changes with criteria.

Task 2: Privacy and Sensitive Data Handling

Prompt:

1. Create a program to store student details like roll number and email.
2. Add marks for each student and display them.
3. Mask sensitive data like email for privacy.
4. Ensure data is not fully visible to unauthorized users.
5. Explain importance of privacy and ethical responsibility.

Code:

```
somes > lab22.py > ...
1  students = [
2      {"roll": 1, "email": "rahul@gmail.com", "marks": 85},
3      {"roll": 2, "email": "ram@gmail.com", "marks": 90}
4  ]
5
6  for s in students:
7      masked_email = s["email"][:2] + "***@gmail.com"
8      print("Roll:", s["roll"], "Email:", masked_email, "Marks:", s["marks"])
```

Output:

```
DIVERSE SELECTION. ( Challenge , 80, 5)
● PS C:\Users\rahul\Desktop\AI Assistanting coding> & C:\Users\rahul\Desktop\AI Assistanting coding\somes\lab22.py
Roll: 1 Email: ra***@gmail.com Marks: 85
Roll: 2 Email: ra***@gmail.com Marks: 90
○ PS C:\Users\rahul\Desktop\AI Assistanting coding> 
```

Explanation:

1. Program stores student academic data.
2. Email is masked to protect privacy.
3. Sensitive information is not fully shown.
4. This follows ethical data protection rules.

Task 3 – AI-Generated Security Risks

Prompt :

1. Create a simple payment processing program.
2. Take card number and amount as input.
3. Show risks like storing card details in plain text.
4. Add validation to prevent invalid inputs.
5. Improve code to make it more secure.

Code:

```
somes > lab22.py > ...  
1 card = input("Enter card number: ")  
2 amount = float(input("Enter amount: "))  
3  
4 if len(card) != 16:  
5     print("Invalid card number")  
6 else:  
7     print("Payment of", amount, "processed successfully")
```

Output:

```
● PS C:\Users\rahul\Desktop\AI Assistanting coding> & C:\Users\rahul\AppData\Local\Microsoft\Windows\AI Assistanting coding\somes/lab22.py  
Enter card number: 1234567890123456  
Enter amount: 500  
Payment of 500.0 processed successfully  
○ PS C:\Users\rahul\Desktop\AI Assistanting coding> |
```

Explanation:

1. Program processes basic payment input.
2. It checks card number length for validation.
3. Storing card details directly is risky.
4. Secure coding is needed to prevent fraud.

Task 4 – Accountability in Human–AI Collaboration

Prompt :

1. Create a traffic signal control simulation program.
2. Change signals automatically based on time.
3. Show how AI can control real-world systems.
4. Discuss responsibility if system fails.
5. Identify who is accountable for errors.

Code:

somes >  lab22.py > ...

```
1  import time
2
3  signals = ["Red", "Green", "Yellow"]
4
5  for i in range(3):
6      print("Signal:", signals[i])
7      time.sleep(1)
```

Output:

```
Payment of 500.0 processed successfully
● PS C:\Users\rahul\Desktop\AI Assistanting coding> & C:\Users\rahul\AppData\Local
I Assistanting coding\somes/lab22.py"
Signal: Red
Signal: Green
Signal: Yellow
○ PS C:\Users\rahul\Desktop\AI Assistanting coding> █
```

Explanation:

1. Program simulates traffic signals.
2. Signals change automatically with time.
3. AI systems can control such processes.

Task 5 – Limitations of AI in Critical Systems

Prompt:

1. Create a program to detect faults in aircraft systems.
2. Take input values like temperature or pressure.
3. Identify abnormal values and show warnings.
4. Explain need for expert validation.
5. Discuss limitations of AI in safety systems.

Code:

```
somes > lab22.py > ...  
1  # AI generated code example  
2  a = int(input("Enter number: "))  
3  b = int(input("Enter number: "))  
4  
5  print("Sum:", a + b)
```

Output:

```
PS C:\Users\rahul\Desktop\AI Assistanting coding> & C:\Users\rahul\Desktop\AI Assistanting coding\somes\lab22.py  
Enter number: 5  
Enter number: 10  
Sum: 15  
PS C:\Users\rahul\Desktop\AI Assistanting coding> 
```

Explanation:

1. This is a simple AI-generated program.
2. Users must understand code before using it.
3. Blind copying can cause mistakes.